

Week 2 (LAB1): Paper Airplanes Design

PSTAT122: Design and Analysis of Experiments

Lily Nishizawa

STUDENT NAME

- Lily Nishizawa (A0A8710)

Due Date

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1 Methods

1.1 Aim

The aim of this lab was to investigate whether a paperclip attached to the nose of a paper airplane will make it fly further.

1.2 Procedure

When it was someone's turn to throw the paper airplane, they stood at the end of the tape measure and threw it in the direction of the tape. Another student stood along the side and watched for the point in which the plane hit the ground. The flight length measurement was taken at this point.

For the throws with the paperclip, everyone placed the paperclip at the front of the plane to the best of their ability. For the second trial, everyone put the paperclip in the same spot.

1.3 Randomization

The order of throws was randomized in order to eliminate bias and increase validity by providing a probability structure.

I utilized R to randomize the order of throws. I first created a vector with all of my group members' names in alphabetical order followed by “_with” and “_without” so that each member would randomly have a throw with and without the paperclip. The code and order for the first trial is as follows:

```
1 trials <- c("Aakash_with", "Aakash_without", "Alisha_with", "Alisha_without",
2             "Andres_with", "Andres_without", "Atia_with", "Atia_without",
3             "Brendan_with", "Brendan_without", "Daniel_with", "Daniel_without",
4             "Grace_with", "Grace_without", "Lily_with", "Lily_without",
5             "Maya_with", "Maya_without", "Raiyan_with", "Raiyan_without",
6             "Ryan_with", "Ryan_without", "Zachary_with", "Zachary_without")
7
8 xxx <- 123
9 set.seed(xxx)
10 ordered_trials <- sample(trials)
11 print(ordered_trials)
```

```
[1] "Lily_with"      "Raiyan_with"    "Grace_without"  "Alisha_with"
[5] "Brendan_without" "Maya_without"  "Daniel_with"    "Andres_with"
[9] "Alisha_without" "Ryan_without"  "Andres_without" "Brendan_with"
[13] "Raiyan_without" "Zachary_with"   "Maya_with"      "Ryan_with"
[17] "Grace_with"     "Aakash_with"    "Lily_without"   "Atia_with"
[21] "Daniel_without" "Zachary_without" "Atia_without"   "Aakash_without"
```

For the second trial, I used the same vector and used a different seed such that the order would be randomized again. The code and order for the second trial is as follows:

```
1 yyy <- 321
2 set.seed(yyy)
3 ordered_trials2 <- sample(trials)
4 print(ordered_trials2)
```

```
[1] "Ryan_without"  "Maya_without"  "Grace_with"     "Lily_without"
[5] "Maya_with"     "Alisha_without" "Lily_with"      "Daniel_with"
[9] "Brendan_with"  "Zachary_with"   "Daniel_without" "Aakash_without"
[13] "Andres_without" "Ryan_with"      "Zachary_without" "Brendan_without"
[17] "Raiyan_with"   "Grace_without"  "Raiyan_without" "Atia_without"
[21] "Andres_with"   "Alisha_with"    "Atia_with"      "Aakash_with"
```

1.4 Confounding Variables

As we took the paperclip on and off our planes, it is possible that the plane shape may have changed slightly; for example, my plane became slightly crumpled as a result of being handled for an extended amount of time. Furthermore, while there was not extremely strong wind at the time of data collection, there was a slight breeze that could also have affected the flight of the planes. In order to minimize this effect to the best of our ability, we opted to fly our planes in a hallway that was slightly isolated from the direct outdoors, though it is possible that the wind still affected the length of the planes' flights. It is possible that the breeze resulted in an increase or decrease in flight length for the planes, and it is also possible that this effect differed between throws.

In our attempts to minimize bias due to plane model, we all made our own paper airplanes, which as a result created a possibility that the paperclip placement on the planes differed from person to person. In addition, because each member threw their own plane, it is possible that the force and angle which the planes were thrown at differed from person to person, but also within one person as well. It is possible that group members threw the plane differently with/without the paperclip, but also between trials.

Furthermore, it is likely that our measurements were not exact as the planes (1) did not always stop when they hit the ground and (2) did not fly exactly above the tape measure, resulting in us having to estimate (to our best ability) how far the plane had flown. Sometimes the planes flew past the end of the tape measure, which also made further measurements more difficult and possibly less accurate.

Due to the fact that our sample size was not very large, we opted for two trials to maximize the validity of the experiment within our time restraint and resources. However, the sample size remains to be small, and therefore could result in low generalizability.

2 Results

2.1 Raw Data, Two Trials

```
1 library(readxl)
2 library(knitr)
3 plane_data1 <- read_excel("Lab 1.xlsx", sheet = "Sheet1")
4 kable(plane_data1, col.names = c(" ", "Trial 1", " ", " ", "Trial 2", " ", " "),
5       caption = "Plane Throw Lengths (in)", align = "l")
```

Table 1: Plane Throw Lengths (in)

	Trial 1			Trial 2		
Name	Without	With	Difference	Without	With	Difference
Aakash	391	340	-51	218	284	66
Alisha	125	121	-4	193	170	-23
Andres	60	192	132	156	132	-24
Atia	87	104	17	114	101	-13
Brendan	117	138	21	39.5	132	92.5
Daniel	187	172	-15	133	220	87
Grace	91	216	125	195	228	33
Lily	138	276	138	192	195	3
Maya	83	189	106	198	222	24
Raiyan	132	156	24	120	149	29
Ryan	220	216	-4	190	191	1
Zachary	34	252	218	114	144	30

2.2 Raw Data, Trials Combined

Combining the two trials, the table is as follows (“2” marks Trial 2)

```

1 plane_data2 <- read_excel("Lab 1.xlsx", sheet = "Sheet2")
2 kable(plane_data2, caption = "Plane Throw Lengths (in)", align = "l")

```

Table 2: Plane Throw Lengths (in)

Name	Without	With	Difference
Aakash	391.0	340	-51.0
Aakash 2	218.0	284	66.0
Alisha	125.0	121	-4.0
Alisha 2	193.0	170	-23.0
Andres	60.0	192	132.0
Andres 2	156.0	132	-24.0
Atia	87.0	104	17.0
Atia 2	114.0	101	-13.0
Brendan	117.0	138	21.0
Brendan 2	39.5	132	92.5
Daniel	187.0	172	-15.0
Daniel 2	133.0	220	87.0
Grace	91.0	216	125.0

Name	Without	With	Difference
Grace 2	195.0	228	33.0
Lily	138.0	276	138.0
Lily 2	192.0	195	3.0
Maya	83.0	189	106.0
Maya 2	198.0	222	24.0
Raiyan	132.0	156	24.0
Raiyan 2	120.0	149	29.0
Ryan	220.0	216	-4.0
Ryan 2	190.0	191	1.0
Zachary	34.0	252	218.0
Zachary 2	114.0	144	30.0

2.3 Summary Statistics: Without, With, Difference

```

1 sum_stats_data <- plane_data2[sapply(plane_data2, is.numeric)]
2 sum_stats <- data.frame(
3   n = sapply(sum_stats_data, function(x) sum(!is.na(x))),
4   Mean = sapply(sum_stats_data, mean, na.rm = TRUE),
5   SD = sapply(sum_stats_data, sd, na.rm = TRUE),
6   Min = sapply(sum_stats_data, min, na.rm = TRUE),
7   Max = sapply(sum_stats_data, max, na.rm = TRUE)
8 )
9 kable(sum_stats, align = "l", caption = "Summary Statistics")

```

Table 3: Summary Statistics

	n	Mean	SD	Min	Max
Without	24	146.9792	75.18130	34	391
With	24	189.1667	59.89024	101	340
Difference	24	42.1875	65.09972	-51	218

2.4 Hypothesis Testing

2.4.1 Defining Variables and Hypotheses

Null Hypothesis $H_0 : \mu_D = 0$

Alternative Hypothesis $H_1 : \mu_D \neq 0$

Where:

- μ_w is the mean flight length with a paperclip
- μ_{wo} is the mean flight length without a paperclip
- μ_D is the difference of the means of flight length between planes with and without a paperclip ($\mu_D = \mu_w - \mu_{wo}$)

2.4.2 Test and Results

Below are the code and results of the hypothesis test.

```
1 ttest_res <- t.test(plane_data2$Difference, alternative = "t", mu = 0)
2
3 ttest_table <- data.frame(
4   "t-statistic" = ttest_res$statistic,
5   "df" = ttest_res$parameter,
6   "p-value" = ttest_res$p.value,
7   "95% CI Lower" = ttest_res$conf.int[1],
8   "95% CI Upper" = ttest_res$conf.int[2],
9   check.names = FALSE
10 )
11
12 kable(ttest_table, align = "l", caption = "Hypothesis Test Results")
```

Table 4: Hypothesis Test Results

	t-statistic	df	p-value	95% CI Lower	95% CI Upper
t	3.174755	23	0.0042258	14.6983	69.6767

2.4.3 Conclusion

The p-value of 0.0042258 is less than the significance level of $\alpha = 0.05$, leading us to reject the null hypothesis that there is no difference between the two conditions. These findings provide evidence that there is a statistically significant difference between the two conditions, with and without a paperclip.

Additionally, the 95% confidence interval suggests that the mean difference falls between 14.70 and 69.68 inches, which does not include 0, providing further evidence that the paperclip may have a meaningful impact on the flight of the paper airplane.

3 Discussion

3.1 Assumptions

3.1.1 Independence

The differences between paired measurements in a t-test should be independent. In this experiment, each student is independent of each other, so this assumption seems reasonable.

3.1.2 Normality

Since the sample size in this experiment is small ($n = 24$), the distribution of differences should be normal (or at least approximately normal). This assumption may not hold, given the sample size is too small to apply CLT.

3.1.3 Continuity

The data in this experiment should be continuous, which holds in this experiment, since the flight distances are a continuous variable measured in inches.

3.1.4 Pairing

Since each student had both treatments of “with” and “without” paperclip, this assumption holds.

3.2 Drawbacks

One limitation of this study is the small sample size and limited number of recorded throws per student. Although practice throws were allowed, only two measured trials per condition were used in the analysis, which may not fully capture the natural variability in flight distance.

Another drawback is potential inconsistency in airplane construction and paperclip placement. While students were instructed to place the paperclip in the same position each time, small differences in placement and slight changes to the airplane’s shape over repeated throws could have affected flight distance independently of the treatment.

Measurement error is also a concern. Distances were estimated based on where the airplane first hit the ground in the direction of the tape measure, which may have been difficult to judge precisely, especially when planes did not land directly above the tape or traveled near the ends of the measuring range.

Finally, variability in throwing technique remains a limitation. Differences in force, angle, and release height—both between students and across throws for the same student—could have influenced the recorded distances and added variability not attributable to the presence of the paperclip.